

A Web Based Cardiac Care Information System



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Declaration and Approval

We declare that this work has not been submitted to Strathmore University or any other University for the award of a Diploma in Business and Information Technology or any other Diploma.

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Abstract

Patient-centered communication as a philosophy was developed to cater to the medical status and social needs of patients. However, in this particular project the focus will be the facilitation of cardiac care in patients afflicted with cardiovascular diseases. The primary reason will be to ameliorate the state of communication that currently exists between cardiologists and patients suffering from cardiovascular diseases. Research indicates that there is a significant reduction in deaths and declining health status as a consequence of personalized care for the patients who have been afflicted by cardiovascular diseases. This proposal outlines the frameworks, methodology and implementation strategy that promises to significantly improve patient care quality and improve cardiac care communication channels. Through intense literature-backed research, this project endeavours to address the problem of substandard levels of patient-centered communication. The proposed project serves user requirements through the implementation of multiple portals that have diverse user stories and classes. The project represents a comprehensive approach to modernizing patient-centered communication. The developed solution is an upgraded system that uses an agile development approach to ensure that the requirements are fulfilled. The tool used for the development of the web application system is MERN (Mongo DB, Express js, React, Node js) Stack, Artificial Intelligence APIs and Visual Studio Code. The end of this project is noble and worthwhile, and it shall be beneficial to the Cardiac care medical field.

Chapter 1: Introduction

1.0 Background

Patient-centered communication is one of the most accurate harbingers of a successful and well-established health care system. Accurate and effective patient-centered communication can only be achieved by first matching the protocols in place with the health-seeking behaviour of the type of demographic that is to be served. A study done by Oxford Academic on health-care seeking behaviour among East-African patients afflicted with Cardiovascular Diseases (CVD) found that the biggest inhibitors to accessing healthcare services was the lack of information about their disease and the high cost of care. This underscores the need for communication strategies that cater to the proletariat barely scraping by to enhance patient understanding and longevity (Hertz, 2022)

In third world nations in Africa, there is a dearth of knowledge and poor understanding of the potential risks of CVDs or developing CVDs. Beyond the widely known quintessential physiological risk factors, psychosocial factors amongst African nations like anger, depression and anxiety especially among the materially deprived populations has been proposed as an emerging risk factor for CVDs. Due to the rampant poverty in most East African nations, there is also high illiteracy levels. Growing and unplanned urbanization exacerbates an already tragic situation due to improper organization among health care systems in the working class or lower. Existing models of health behaviour suggest that educated individuals that are aware of the associated risks or possibilities of CVDs employ risk mitigation measures that drastically lower their chances of developing the disease (Murunga, Kyobutungi, & Grobbee, 2019).

According to a study done in Nairobi by Kyobutungi (2020), on understanding of perceptions towards CVDs among residents in the informal urban setting, the adult demographic (35 years and above) tends to be the most affected by CVDs. A key finding of this work was the discovery of the astonishing levels unawareness and ignorance of the risk factors of CVDs that were as a consequence of the widespread poverty, illiteracy and ignorance. Purchasing power is one of the biggest reasons for most problems in any domain, let alone health. When the aristocracy catches a cold, as it is said, the working class dies of pneumonia. Lifestyle decisions are influenced a majority of the time by social class in most East African countries. Majority of the respondents in the study did not see the direct correlation between habits like smoking and excessive alcohol

consumption and CVDs. Furthermore, the respondents also did not understand the link between strokes and heart attacks with Alcohol consumption, smoking and insufficient physical activity. It was clear that the respondents lacked the knowledge and awareness to see the interrelationship between said factors and CVDs (Murunga, Kyobutungi, & Grobbee, 2019).

A study done by The American Journal of Managed Care (AJMC) (2020) , among patients suffering from Heart Failure (HF), it was demonstrated that excellent patient-centered communication with their primary care-givers is associated with a 30% lower risk of death, independent of age, sex, marital status, education and health literacy (Fabbri, 2020)

In Kenya, National guidelines for most CVDs and Non-communicable Diseases risk factors have been developed. Through Kenya's NCD strategic plan 2021-2025, the Ministry of Health envisioned to establish mechanisms to integrate NCD prevention and control in government sectors at a country and national level, promote healthy lifestyles and interventions to address the controllable risk factors of NCDs and CVDs like physical activity, tobacco and Alcohol consumption, facilitate and conduct research and surveillance for the control of the spread of NCDs and CVDs, establish and maintain effective monitoring and evaluation systems for NCDs and CVDs among many other reasons. The Kenya Health policy streamlines the direction the health sector ought to take to ensure that the overall status is in line with The Constitution of Kenya but is significantly refined and improved.

1.1 Problem Statement

It is no secret in nation like Kenya, there exists insufficient and substandard levels of affordable patient-centered communication. Whether literate or illiterate, proletariat citizens of East African nations suffering from CVDs and NCDs have moderate to low access to patient-centered communication. This is due to the purchasing power constraint that afflicts most of the people and also the unequivocally uneven ratio of patients to care-givers. Kenya has an average of 1.6 physicians and 10 per 10 000 of the population in 2018/19. The number of hospital beds was 13.3 and reported as the 'The national average in-patient bed density' by the health facility assessment in 2018/19. In 2018, a locally developed clinical tool that included locally relevant clinical guidelines for CVD prevention, detecting and managing AF was developed to assess CVD risk. However, no clinical registers of people with a history of ARF and RHD were available. In 2010, national clinical guidelines for managing diabetes were developed based on local and international

best practices and updated in 2018. However, no system to measure the quality of care provided to people who have suffered acute cardiac events (Mbau, Nel, & Fourie, 2021).

The problem of insufficient patient-centered communication has led to increased mortality rates amongst the working class of the nation. The backbone of a vibrant and efficient society is the working class thus if they are not catered for, the economy of the nation suffers as a consequence. Members of the bourgeoisie community find themselves even seeking medical care out of the country as the problem of well-established patient-centered systems presents itself. Ineffective communication had been described as the most accurate indicator of worse outcomes amongst patients with limited health literacy. There has been a direct correlation that has been observed between inadequate health literacy and increased rate of mortality and hospitalization (Fabbri, 2020).

There is a stench of hopelessness wafting through the air in a nation that has majority of its citizens unable to afford or have access to inexpensive high-quality care. The consequence of the gap is the continued risk of death, lowered life expectancy of the country's residents and development enclave due to only a small minority of the citizens (bourgeoisie) being able to afford said care. A study published in health policy and planning by Oxford Academic, noted that high expenses in medical care can deter individuals from seeking necessary treatments affecting negatively the health outcomes and overall life expectancy of the populace (Kulka, McKay, & Harman, 2017).

1.2 General Objective

To develop a communication web-based solution to facilitate patient-centered communication in a cost effective and accessible way that eases the connectivity between the patients and care-givers.

1.2.1 Specific Objectives

- i. To analyze how patients currently afflicted with CVD come into contact with care providers such as Cardiologists.
- ii. To review current patient communication-centered technologies and existing challenges that patients face in the current way of communication.
- iii. To design an interactive Web Application using Unified Modelling Language (UML) diagrams and various concepts in System Analysis and Design.

- iv. To develop a web application for patients afflicted with CVD and seek affordable patient-centered communication.
- v. To test the developed web application and make sure it ameliorates the lives of the patients.

1.3 Research Questions

- i. How do patients currently communicate with cardiologists in areas that are not in proximity?
- ii. What are the challenges that the patients are currently facing that they feel exacerbates their already declining health status?
- iii. What are the key functionalities that are needed for an understandable/ User friendly and interactive user interface
- iv. How can the web application meet the gap in patient-centered communication between patients and doctors alike?
- v. How will the beta testing of the app be instantiated?

1.4 Justifications

Research indicates that patient-centered communication is associated with reduced risk of death by 30% among patients afflicted with CVDs. It has been reported in different studies that the needs of the patients are not always met by the primary care-givers. Inadequate and ineffective communication has been popularized among medical circles as the one of the leading factors for worse outcomes among patients with limited health literacy (Fabbri, 2020). It is thus incumbent upon us to develop effective communication channels with the help of emergent technologies to facilitate better communication.

1.5 Scope

The proposed study focuses on solving the issue of communication between doctors and patients while providing an inexpensive channel to facilitate contact. The patients will have the option of using the system themselves or a trained care-provider assigned to them on their behalf. They will have contact with the cardiologists through their user-defined accounts that they will have their pertinent medical records stored and accessed by them and the doctor. Diagnosis and prescriptions too can be accessed by the nurses and the patients.

1.6 Limitations(s) in the project

Some limitations that may affect the project include high level of illiteracy amongst geriatric patients with respect to the operation of Information Technology. Additionally, the project may face challenges related to data privacy and security due to the sensitivity of medical information that people would reluctantly trust an online resource with their own information.

Delimitation(s) of the project

To train the primary care-givers, and patients alike, on the Information system that is to be implemented as the primary channel to facilitate the patient-centered communication.

Chapter 2: Literature Review

2.1 Introduction

This Chapter looks to review current patient communication-centered technologies and existing challenges that patients face in the current way of communication, analyze how patients currently afflicted with CVD come into contact with care providers such as Cardiologists, design an interactive Web Application using Unified Modelling Language (UML) diagrams and various concepts in System Analysis and Design and develop a web application for patients afflicted with CVD and seek affordable patient-centered communication.

2.2 Patient-Doctor communication in Kenya

Due to the dynamic nature of the health and care landscape, there is now more than ever the need to integrate the organization of care with the patient placed at the centre. Quintessentially, care systems are often impersonal and focused on the condition rather than the patient as whole. There is a relationality that needs to be included as a responsibility to care for patients as people, and not just hearts, and this care should extend beyond the boundaries of hospital (Cartledge, Ski, & Foldager, 2023)

The modern approach to the care of patients with CVDs postulates a way to create an education and communication system that engages the patient, provides encouragement for self-care, and allows consistent feedback to maintain optimal health status. The existing systems have care characterized by intermittent bursts of attention. Episodic care for patients is not sustainable in the long-term. It is costly and unscalable. Episodic care leads to the deterioration of the patient's health which in most cases of recurrent acute exacerbations in health, heroic therapies are needed to alleviate the situation. In contrast, continuous surveillance serves to curtail the frequency of these acute episodes (Bove, 2010).

The point of focus for this particular project will be the consistent surveillance of CVD patients. This will be coupled with other important functions such as dissemination of pertinent and helpful information to the patients and facilitation of ease of contact with specialists. The surveillance in itself will have to be provider-initiated as a prerequisite for obvious reasons. Statistically speaking, the ill and unmotivated patient will be less likely to initiate contact and conversation. With the

main reason being the high cost of consultation. Provider-initiated telephone service as an example, has been shown to reduce acute exacerbations in health (Bove, 2010).

According to a study done by Radcliffe Cardiology (2010), some of the guiding rules and regulations that have to be paid attention to facilitate effective patient surveillance are the following:

- i. The patient has to be willing to **actively** participate in the process of **their own volition and accord**. A gentle nudge in the right direction will suffice but the use of force will be nothing but an impediment to the healing process.
- ii. The care-providers have to be **trained** on the proper means of relationality with the patient. The care providers need to be astute at reading verbal and non-verbal cues and infer the health status of the patient from the interaction. Hence giving proper recommendations.
- iii. **Investment** in automated and computerized processes that allow for scalability and sustainability. The system has to at least integrate some form of software processes that can facilitate the large amount of traffic and the complexity of some actions.

However, it is technically impossible to have a high-quality personal relationship with every patient needing care, because there are just too many people. In addition to this, most people are firmly placed in the working-class category. Thus, the high rates of consultation fee of the “greedy” doctors can only be afforded by the minority bourgeoisie class. The expectation of a “rainbows and sunshine” state of affairs in the medical space is simply fantasy. The reality is that the medical industry is most concerned about profitability more than anything else. Hence why it is relatively affluent (Bove, 2010).

2.3 Challenges faced in health surveillance and patient-centered communication

Despite the efforts made to curb low-level patient centered communication which only pays attention to the bottom line as opposed to overall well-being of the patient, there are still some limitations that are barriers to progress.

For starters, Telephone interviews that were introduced as a solution for interpersonal communication with the patient, ended up being too costly and time-consuming. Furthermore, the specialists determined there was a limitation in relationship development and depth of narrative due to the nature of the generation of the patients who were mostly used to in-person conversations

(Jennings, Egert, & Fisher, 2023). Having costly solutions to a costly problem only multiples the degeneration. The best solutions are cost-effective and resourceful.

Proper and formidable healthcare systems cannot be run on expensive constituents. The consequence of expensive means of patient-centered communication is the inevitable fallout of most working-class patients forced to have a quasi-batch processing system for their CVDs. The patients will be forced to engage in episodic care as they are forced to save up money and only see the doctor when something worse happens. Keeping in mind that this could have been prevented altogether by simply just having a sustainable and cost-efficient method of surveillance.

Attempts to try and solve the problem of expensive means of contact have been made but to little and moderate success. Again, we have to keep in mind that its in the best interest of the medical industry to keep things expensive for their own profits. There is no incentive for them to create cost effective systems. Medicine is a business, albeit a very lucrative one. It is only people who do not have a piece of the pie that are most likely to address the inequalities and the injustices of the system.

This is the reason why Health Insurance companies exist to try and subsidize the cost of care amongst their subscribers. Yet again we run into another problem, they are incentivized by profits too! Out of the frying pan into the fire it seems. People end up living way below their means as a consequence of ridiculous health expenses because no matter who you are, every single person has or has had a member in their family close or distant who has suffered a chronic disease. If you haven't had one, it will happen sooner or later. Its just the condition, some may say curse, of being human. The development of internet-based communications has been developed to reduce reliance on direct telephone contact or actual physical travelling to the clinic or hospital. Internet-based systems provide sustainable frameworks for personalization and education while circumventing the problem of costly direct physical contact with healthcare provider. Although, little investment into the architecture of such systems has been done again due to the "Profitability" constraint. The main gap presents itself as the substandard development of existing web-based access systems that do not integrate with most of the emergent technologies (Cartledge, Ski, & Foldager, 2023)

2.4 Review of Existing systems

Here are some examples of web-based systems that are to be analyzed to find the inner workings and identify shortcomings that can be rectified as a consequence.

2.4.1 Aga Khan Hospital – Cardiac care

The Aga Khan Hospital has a Cardiac care centre within the hospital that has “attempted” to develop web-based static page for their cardiac unit. They have a static page that is not responsive and has only read-only properties. There is no personalization nor patient centered communication. It is just a page to showcase information on what they do at the hospital. There are buttons at the bottom of the page that allow the following:

- i. One is able to refer a patient through a link at the bottom of the page that leads you to an application form. Although, this requires that one knows their underlying condition before referral.
- ii. One is able to book an appointment for themselves and they are given a security code assigned to their appointment for data privacy.

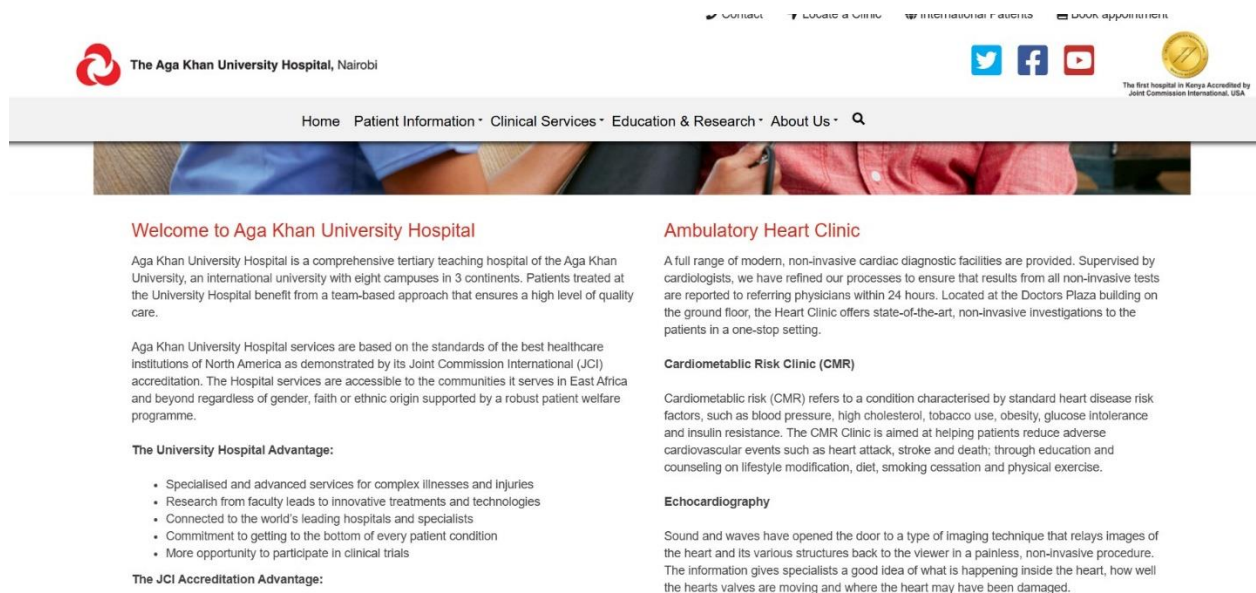


Figure 2.0.1: Aga Khan Hospital Web Application User Interface (Hospital, 2025)



[Contact](#)
[Locate a Clinic](#)
[International Patients](#)
[Book appointment](#)






[Home](#)
[Patient Information](#)
[Clinical Services](#)
[Education & Research](#)
[About Us](#)

Kindly note that all personal, sensitive and health data collected shall be processed in accordance with the Data Privacy Act No 24 of 2019. For more information on your rights and responsibilities, please check our data privacy policy here, or request for a data privacy pamphlet at your clinic. [Read our Data Privacy Statement.](#)

Book an Appointment

First name *
Middle name
Last name *

Phone number*
Email address *

Gender*
Language preference*

Address *
Country of residence*

Preferred date of appointment*
Specialty*

Who are you seeking care for? *

Primary medical issue: *

Figure 2.0.2: Aga Khan Hospital Appointment Booking Form (Hospital, 2025)

2.4.2 Karen Hospital-Cardiac Unit

The Karen Hospital has one of the biggest cardiac centres in the country that offers the best cardiac attention but at a price of course. They too have a webpage that describes the services they offer at the clinic. However, when one clicks on the link to book an appointment, the button leads you to an error 404 page describing that the online booking service is currently not offered. Hence, to book an appointment one either has to call or go directly to the Hospital headquarters in Karen.

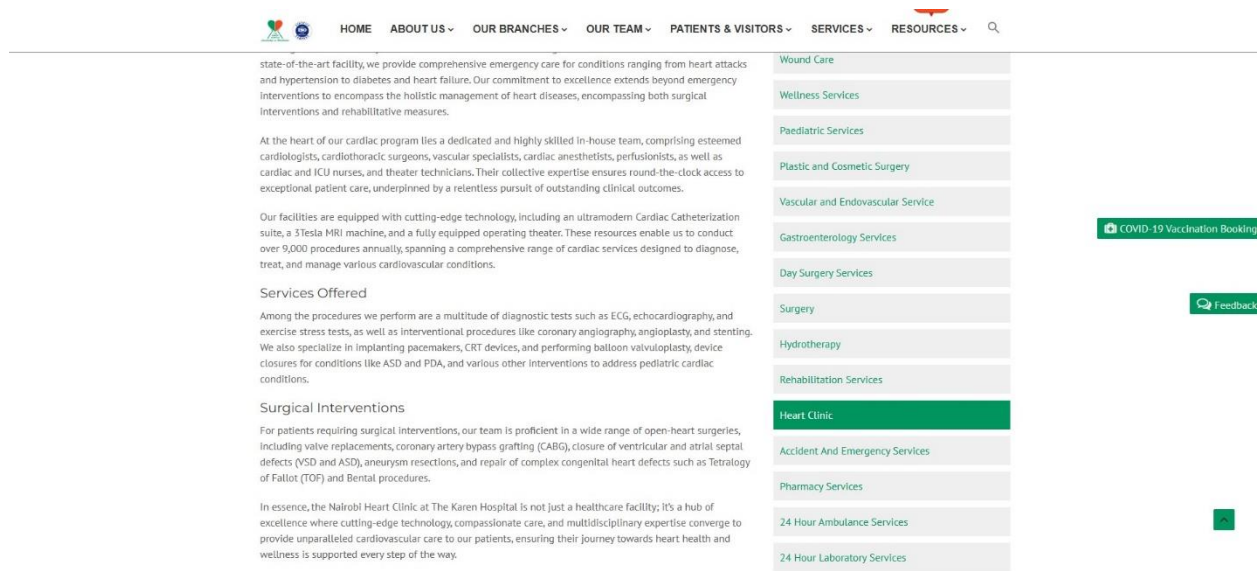


Figure 2. 0.3: Karen Hospital Web Application User Interface (Karen, 2025)

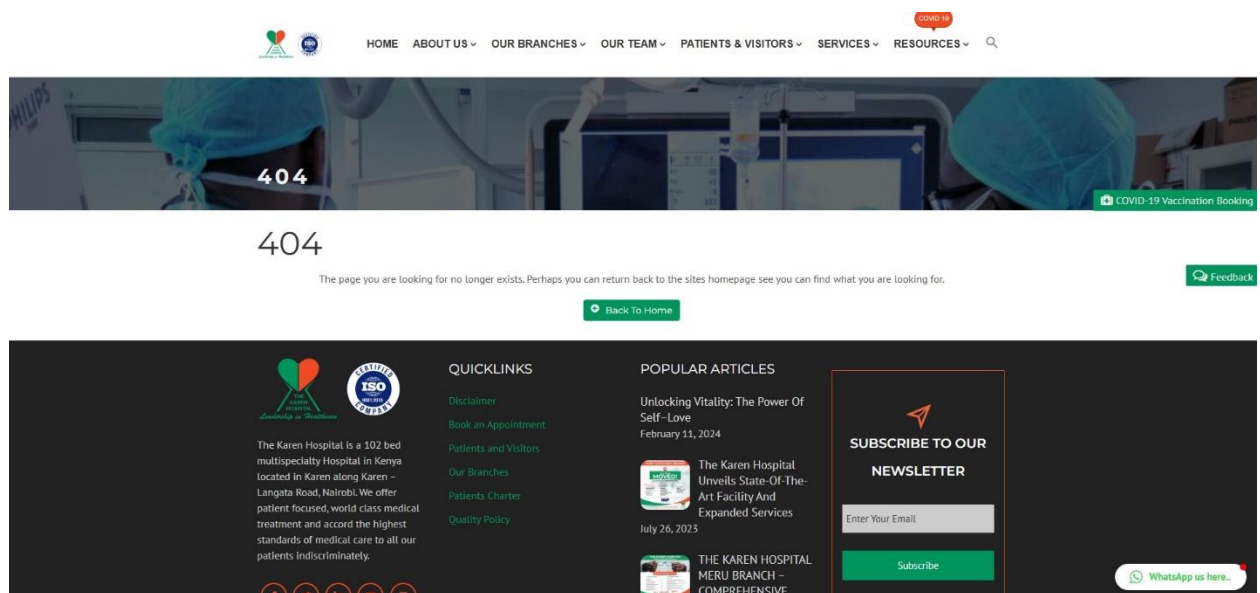


Figure 2. 0.4: Karen Hospital 404 Error page (Karen, 2025)

2.4.3 SpecialistCardiology.com

Specialist Cardiology is a highly regarded cardiology practice located in the Sydney Adventist Hospital campus that had a landing page developed to advertise their practice. The user Interface is fairly simplified to a few buttons that are large, understandable and readable. It has the most understandable and simple User Interfaces of the three that have been picked for investigation. The simplicity is certainly a strength as one of its many attributes (Add citation).

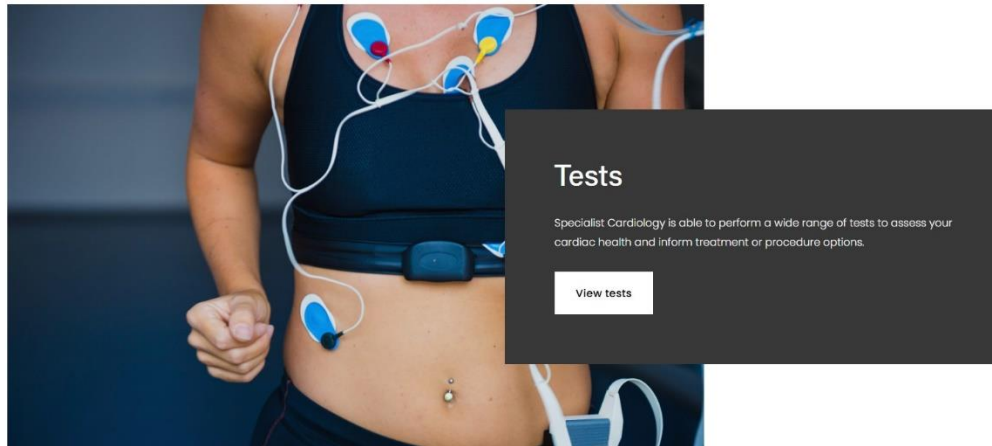


Figure 2.0.5: Specialist Cardiologist Web Application User Interface (Specialist, 2025)

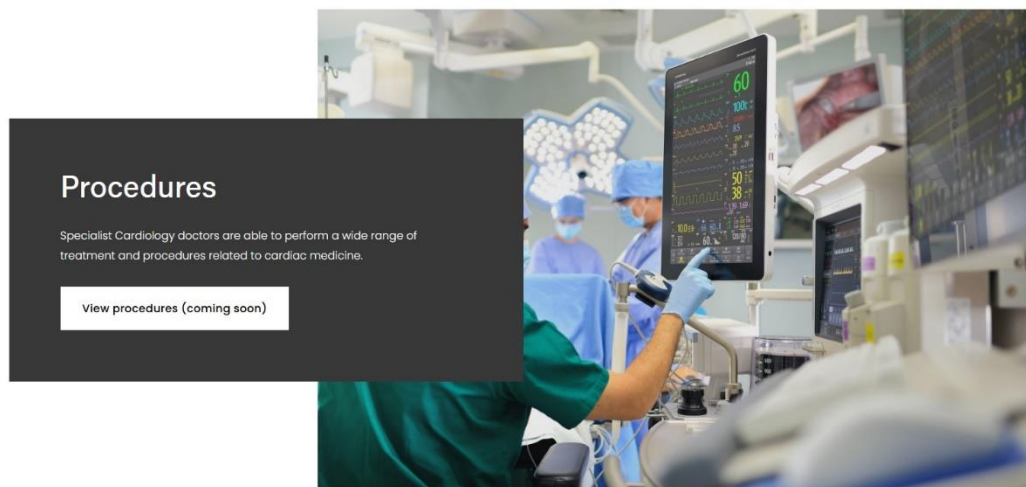


Figure 2.0.6: Specialist Cardiology User Interface (Specialist, 2025)

2.5 Gaps in the existing systems

In the systems picked for investigation, there are certain weaknesses that have been identified that can be rectified with applied knowledge and emergent technology. In the examples above, the following are the gaps that have been identified that can be rectified:

- i. The web Applications presented mostly in Kenya, lack the necessary interactivity that makes the user experience easier and faster. Key functionalities have been omitted leaving no option for the user to settle for the number given to call for appointment booking.
- ii. The information given on the webpage serves to advertise their services offered and not educate on the different topics related to cardiac care. There are no articles or newsletters written by the cardiologists themselves to serve as a tool to ameliorate the lives of the affected.
- iii. Some of the Web Applications are outdated and are unused. The buttons lead to error 404 pages due to lack of an implemented robust and consistent system of updates.

2.6 Conceptual Framework

The conceptual framework below represents Cardiac Care Information System whose main purpose is to facilitate patient-centered communication. The system consists of a patient portal, User management portal, Healthcare Provider Portal and the Core system that contains the database. Patient portal allows the patients to upload medical documents, view doctor profiles, book appointments and submit reviews. User Management portal which is run by the Admin, allows the Admin to monitor the rating system, control access control and manage user registration. The Healthcare Provider Portal that has the cardiologists and nurses allows the cardiologist to manage the prescription system, to view patient records, upload medical guidelines and to update work status. The nurses can view patient records, update work statues and use Artificial Intelligence Symptom checker. The Core system manages medical records, manage the authentication system and finally maintains AI symptom checker.

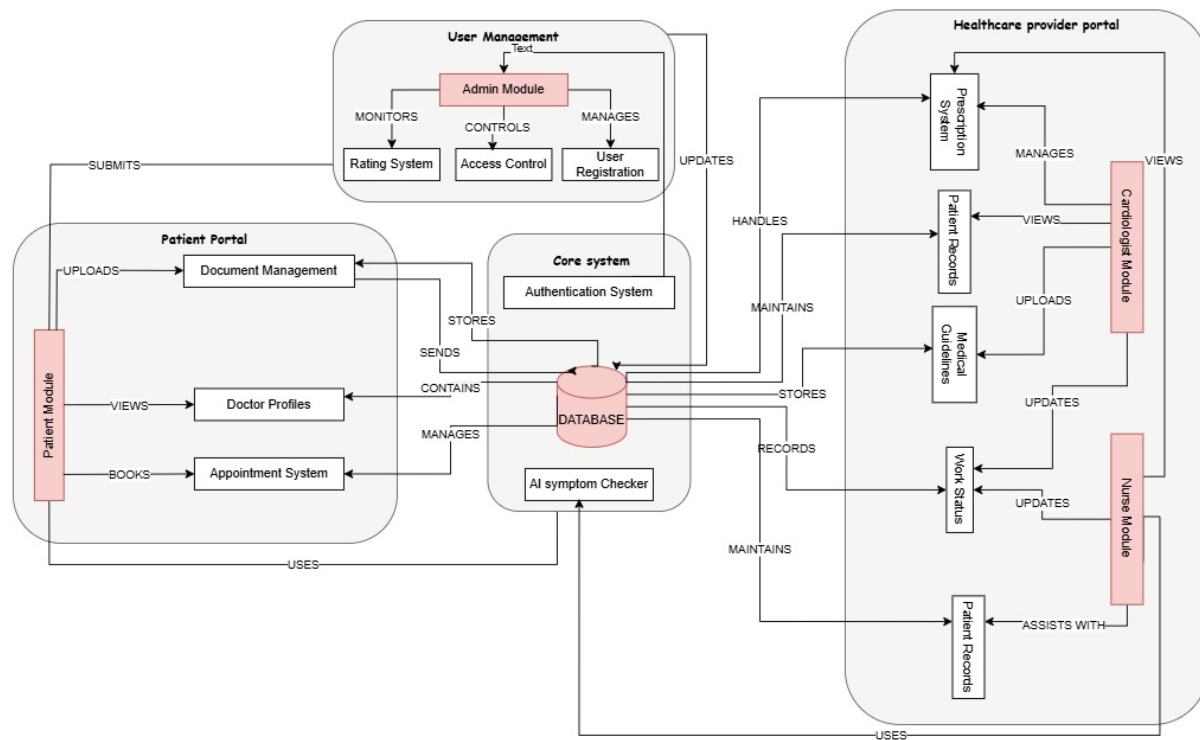


Figure 2.0.7: Conceptual Framework for the Cardiac Care Information System

CHAPTER 3: Methodology

3.1 Introduction

Different methodologies have been analyzed and recommended in this section; data types have been defined in accordance with the means of data collection that will be used for the data. Analysis and Design diagrams have been drawn to illustrate the best approach to the system's instantiation.

3.2 Research Paradigm

For this particular project, Object Oriented Analysis and Design Method (OOADM) will serve the needs of the Information system design best compared to Structured System Analysis and Design Method (SSADM). This conclusion is due to the following reasons:

- i. OOADM mirrors real-life objects such as in this case nurses, patients and cardiologists. The objects have defined user stories (e.g. Nurses log patient vitals) that are all different. Since the system has multiple user types, OOADM is best suited for this kind of system.
- ii. OOADM has objects and classes that have both data and functions that are specific to each class or object. In a Cardiac Care Information system having objects with their own data and methods is better than modules containing functions.
- iii. OOADM has loosely coupled objects that are independent and only interact when necessary, which is applicable to a Cardiac Care Information System that wants to facilitate patient centered communication.
- iv. OOADM supports agile methodologies such as Scrum that allow for new features to be deployed quickly.
- v. OOADM because of the loosely coupled components, modules do not require other ones to finish before the next one starts which leaves room for high scalability.
- vi. Due to the complexity of the system where each user has different functions, reusable parts such as objects or classes are best suited for such a system.

3.3 Methodology

Agile Software Development Methodology was the targeted Methodology this project intends to use to implement said project. Particularly, Scrum would be the most ideal for Cardiac Healthcare system. Scrum, technically speaking is considered a framework and not a process or a methodology.

Scrum Agile Software Development Methodology

Scrum as a framework is used to address complex adaptive problems while self-organizing tasks and working towards a common goal. As far as management frameworks go, Scrum definitely has clear advantages over the rest of the frameworks such as:

- i. It is simple to grasp and understand.
- ii. There is an emphasis on self-organization
- iii. It is very lightweight

Scrum involves a methodology that is often referred to as the Sprint. A sprint is an effort that is geared towards product development that lasts for a period of no more than a month. A Sprint, as the name suggests, is relatively short and includes a well-articulated goal the designers aim to achieve. A new Sprint is instantiated as soon as a Sprint ends and are repeated until the product development effort comes to an end. The following are the stages involved in the Scrum process

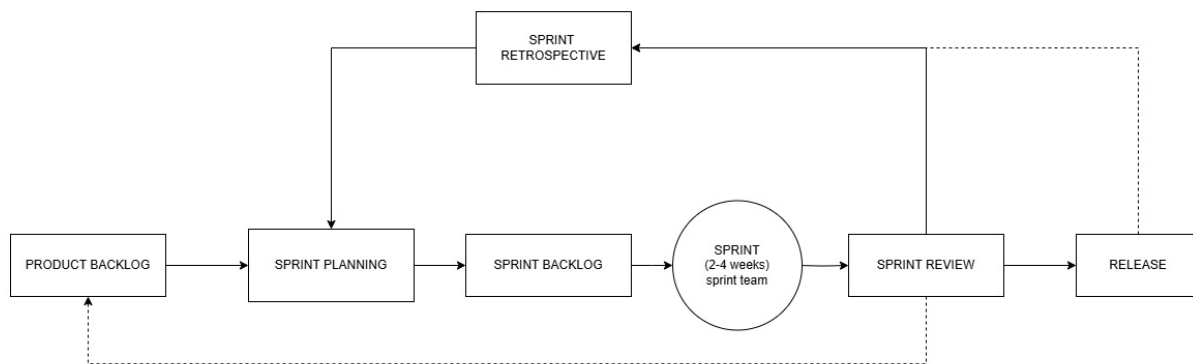


Figure 3.0.1: Sprint Agile Development Model

Sprint Review: If the product has yet to have all its features achieved it will be checked at this stage and proceeds to the Sprint Retrospective Stage.

Sprint Retrospective: Quality level of the product is evaluated at this stage.

Product Backlog: The product is organized according to the prioritized features.

Sprint Backlog: There are two parts to this stage that involves the product getting assigned features to Sprint and the Sprint planning meeting.

3.3.1 Data Gathering techniques

The project problem will be analyzed through requirement gathering techniques, that includes the reviewing of existing systems, stakeholder interviews and a bit of observation. Various types of data will be collected to ensure the needs of the users are met.

Data Types

The following are some of the data types that will be collected:

- i. Patient data: Medical history, ECG readings, vital signs, demographic details
- ii. Nurse-reported data: Symptom tracking, emergency alerts and pharmaceutical administration facilitation.
- iii. Cardiologists' data: Treatment recommendations, offered remote assessments.
- iv. System Feedback: Nurses and Cardiologists give their feedback regarding the system

Primary Data collection methods (qualitative and quantitative):

- i. Electronic Health care records will be used to determine the necessary metrics for cardiac care.
- ii. Interviews and surveys to provide key insights from the nurses and the doctors on inefficiencies that can be rectified.
- iii. Observation of real-time interactions between patients and doctors.
- iv. Research: reading and inferring of articles and studies such as the one done by the European Heart Journal on the Data Standards for Heart Failure (2022).

Purposes for Data Collection

The data that will be collected will be used for the following purposes:

- i. System Design: Information system designed has to be in line with current and updated modern medical workflows

- ii. Ease the decision-making process of the cardiologists and nurses through the provision of timely, readily available and relevant information.
- iii. Through evaluation and analysis of improved or declining patients' outcomes we can then accurately assess the effectiveness of the system
- iv. Ensuring system remains adaptable to future needs by assessing current, past and projected needs.

3.3.2 Analysis and Design

For this particular project, Object Oriented Analysis and Design Methodology will be used with Agile sub-methodology facilitated by the Scrum framework. The Agile/Scrum methodology will be used for the following reasons:

Distributed System Architecture

Since it will be a web-based application, it will have remote access requirements, multiple interfaces and real-time data synchronization.

Management of Change

Evolution of health requirements creates the need for frequent updates and improvements. This will be coupled with regulatory compliance updates that improve the ability to adapt to user feedback.

System Complexity

Multiple user types are involved in the system as a consequence there are complex interactions between the modules. Real time data handling requirements exacerbates the need for scalability and maintenance.

Diagrams

The diagrams that will be used to illustrate how the system will work will be a combination of Class diagrams, sequence diagrams and Use case diagrams.

- i. Primary: Class diagrams map real-world entities into software objects, easily maps into modern web frameworks and supports object-oriented implementation.

- ii. Secondary: Sequence Diagrams show the flow of messages between objects over time and will be used to illustrate API interactions.
- iii. Initial phase: Use case diagrams will be used for requirements specification and identify system boundaries. Primarily used for Stakeholder communication.

3.3.3 Web-based Cardiac Care Information System

The system that will be built in this particular project is a web-based Cardiac Care Information System. Ideally the project will involve the use of a MERN Stack. Mongo DB will be used for database, Express js and Node js will be used for backend while React framework will be used for the front end. A MERN Stack will be the most efficient way to go as it supports the use of emergent web-application technologies such as the use of Artificial Intelligence APIs. The programming language of choice will be JavaScript which is simple to understand and use for the construction of web-based applications systems. For the Integrated Development Environment (IDE) this project will use Visual Studio Code. Visual Studio Code will be used as it is extremely lightweight and has all the needed extensions ready just at the click of a button. Anthropic API will also be used for the AI symptom checker.

No	Language/Framework	Function
1	React - JavaScript	Frontend
2	Express js & Node js - JavaScript	Backend
3	Mongo DB - Database	Database
4	Anthropic API	Artificial Intelligence API

Table 1: Summary table showing tools to be used for construction

3.3.4 Confirmation Testing

Solution Validation

Here are some of the steps that will be undertaken to confirm that the Cardiac Care Information System meets the intended requirements:

- i. Requirement validation: Initial system requirements and use cases will be used as benchmarks for the features in the Information System.
- ii. Prototype testing and Iteration: A Potentially Shippable Product will be deployed in a controlled environment to allow for incremental improvements before full deployment.

Types of tests required

Three main types of tests will be done to ensure that the Information System satisfies user requirements. Functional testing to ensure core features are delivering according to use cases, Security testing to ensure that data privacy and security is observed, User Acceptance Testing to ensure real world usability is achieved.

Testing tools

Google lighthouse will be used for usability testing as it analyses performance and accessibility. OWASP ZAP will be used for penetration testing to ensure security and data privacy.

3.4 Deliverables

This section entails promised outcomes to be delivered.

3.4.1 System proposal

A system proposal documentation will be provided of the Cardiac Care Information system to clearly identify the various processes following its predicted execution at each stage.

3.4.2 Patient module

A web platform will be tailored to the patients liking allowing them to perform tasks such as uploading of medical files and accessing of AI symptom checker.

3.4.3 Healthcare Provider Module

The web platform will be tailored to the cardiologists and nurses allowing them to perform tasks such as evaluation of patient files, diagnostics, accessing of AI symptom checker and update their work status.

3.3.4 User management Module

The admin operates to mainly manage the user accounts of the pertinent users of the system while ensuring login and authentication protocols are followed

3.4.3 System Documentation

A document holding all relevant information about the Web based Cardiac Care Information System and its development shall be provided.

CHAPTER 4: System Analysis and Design

4.1: Introduction

This chapter entails the different components of the system detailing the functional requirements and non-functional requirements and will also give you a guide on how the system development will be carried through the use case diagram, sequence diagram, entity-relationship diagram, class diagram and the database schema.

4.2: System Requirements

The functional and non-functional requirements prove to be essential in the functionalities required in the system.

4.2.1: Functional Requirements

Some of the functional requirements are stated below by checking the system database, the screen displays the main dashboard.

i. Authentication

This requirement is essential for administrators, doctors and patients. In order to access the system's functionalities, the system requires specific inputs, which is a user's unique email and their password. This will go hand in hand with an encrypted password (OTP) which will be encrypted by bcrypt. The users key in the password in the appropriate field to gain access to the system.

For the administrators, they will need to key in their credentials, that being their email and password. Upon validation by checking the system database, the screen displays the main dashboard of the administrator.

ii. Booking of an appointment

The patient can book an appointment on their own behalf to a doctor of their choosing depending on the speciality of the doctor/cardiologist. The patient can then view their appointments and pay online or decide on cash. The slots for booking are of course omitted as soon as the slot is booked. The appointment is then forwarded to the admin and doctor dashboard.

iii. Doctor Login

Doctor can view the appointments made and the status of the payment of the appointment. There are actions that can be taken on the appointments by the doctor. For instance, once the doctor is finished with the appointment they can complete the entry by clicking on the tick and the button

will read completed. They can also cancel the appointment if needed which will then reflect on the admin dashboard as well.

iv. Management of bookings

Administrators serve to oversee the appointments booked and can add doctors to the system. The administrators manage the credentials of the doctors and the availability of the doctors through their own admin dashboard.

4.2.2: Non-Functional Requirements

The non-functional requirements are stated below:

i. System security and safety

To ensure the security of user data against unauthorized access and malware, the system employs one-time password (OTP) as a form of two-factor authentication. Initially, the user needs to key in their email to acquire the one-time password that is valid for a brief period. After the user has keyed in the one-time password, the system verifies it against the database. The process links each user account to a unique email, preventing the creation of fake accounts.

ii. System Performance

The system is designed to respond quickly to user actions. This is achieved through a real time database that supports offline functionality by catching data on the users device and synchronizing it when the device reconnects to the internet. This ensures fast, real-time data access, enabling swift data driven actions by user

iii. Usability

Ease of use makes the students adapts, ensuring the system is both learnable and memorable. It incorporates features such as user control to provide familiarity to users. Simple navigation guide users through their desired actions. Additionally, the system is interactive, offering feedback on actions and seeking confirmation for sensitive tasks like deletion or cancelation.

iv. System Efficiency

The system is designed to be always accessible to users. Users are logged out if the screen is on standby for a long period of time. When the application is launched it checks for a current user and, if found, directs them to the main activity of the application.

v. System Maintainability

The system is designed for minimal maintenance. It is well-documented, built with simple code and tested frequently. This allows for quick fixes and performance improvements.

4.3: System Analysis and Design

This section demonstrates how the requirements are met. Using Diagrams, they describe the functional requirements in the system. These diagrams include use case diagrams, sequence diagrams, entity-relationship diagram, class diagram, database diagram and wireframes

4.3.1: Use Case Diagram

The figure below shows how the actors interact with the system. The system has 3 actors namely the cardiologist, nurse, patient and lastly the admin. The 3 actors, nurse, patient and the admin share a use case such as login. The use cases provided by the cardiologist includes writing prescriptions to patients, upload articles, manage work status, view their own profiles, view documents, upload documents and schedule appointments. The CRUD Functionality is clearly shown as the administrator can login, manage ratings, manage users, which means they can add user or delete users from the system.

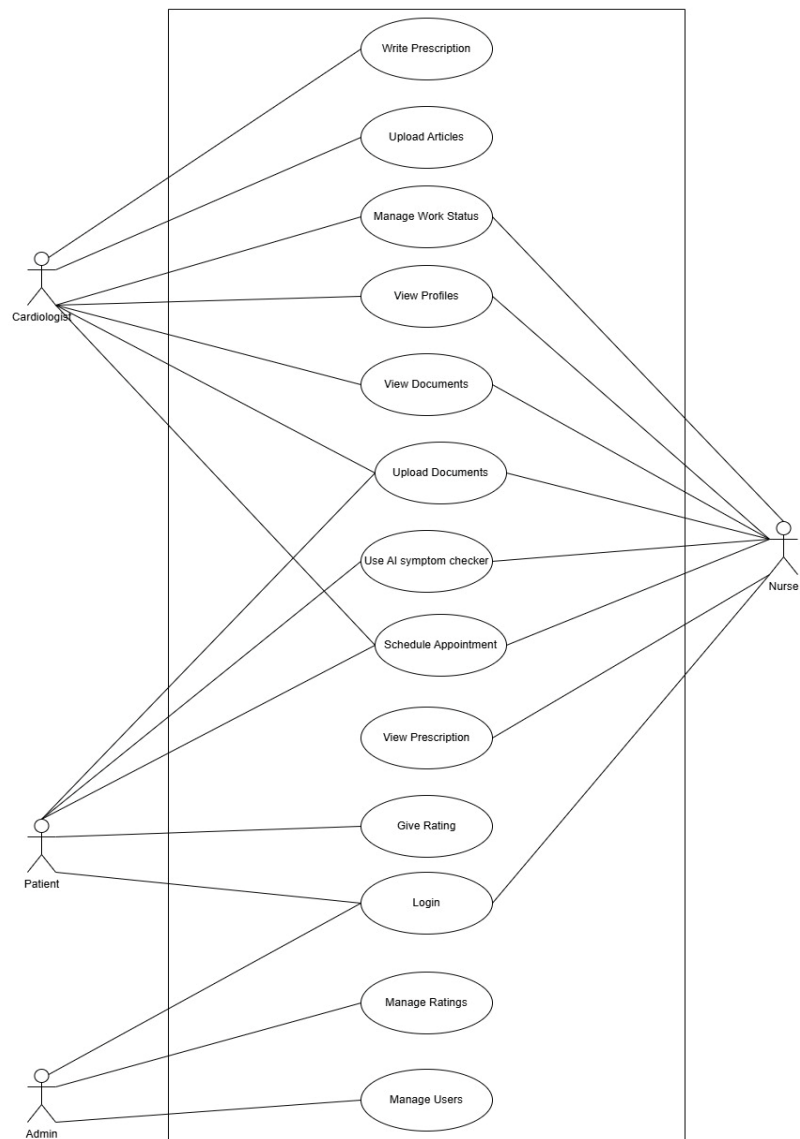


Figure 4.3.1 Use Case Diagram

4.3.2: Sequence Diagram

A sequence diagram depicts the action flow among the system components. In this case given the minimal activity within the demo system for the model, it can be used as a visualization tool between external and internal elements of a system. The figure below shows the flow of communication in detail.

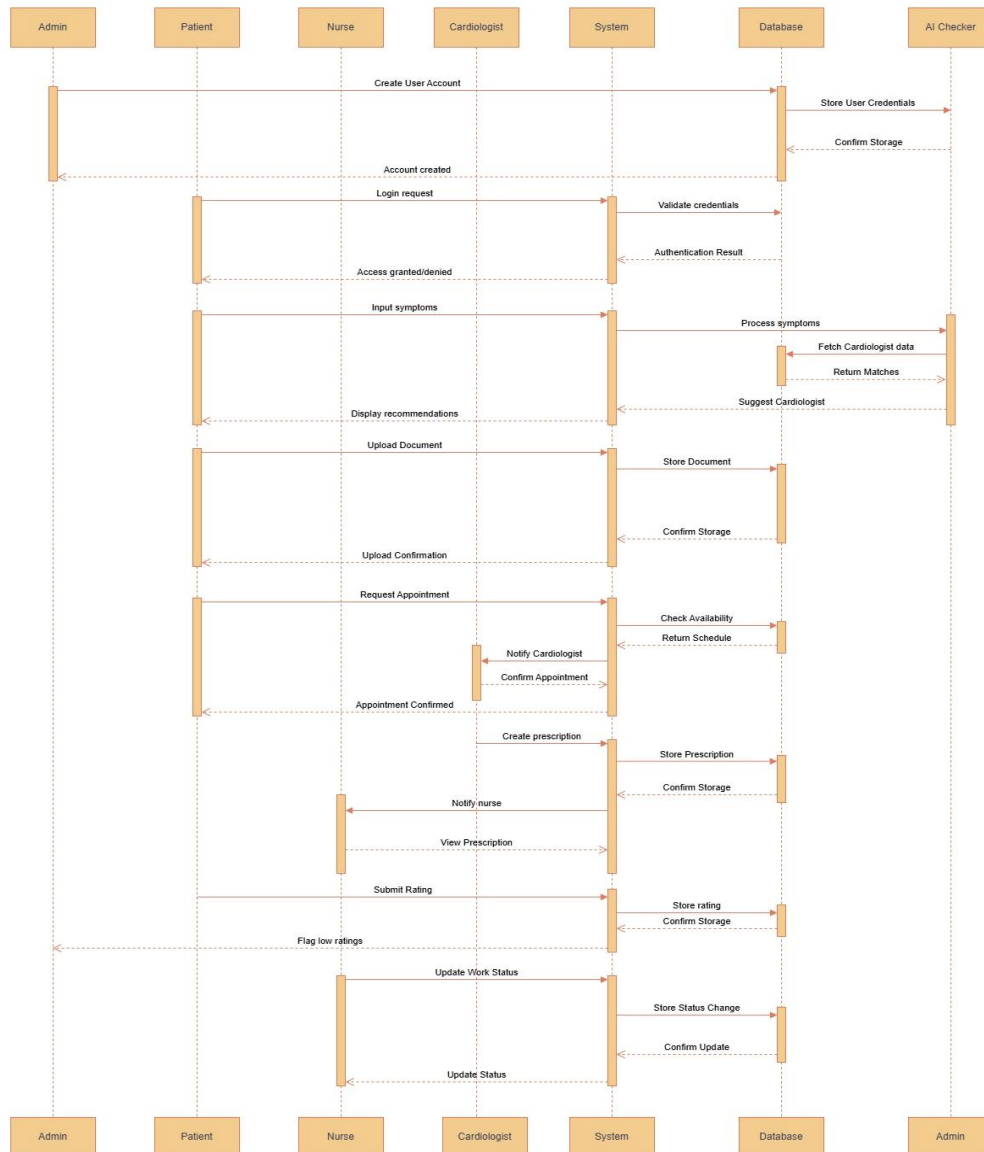


Figure 4.3.2 Sequence Diagram

4.3.3: Entity-Relationship Diagram

An Entity Relationship Diagram is a graphical representation of entities, attributes, and relationship within a database system. It is used to design the database structure by illustrating how data is interconnected.

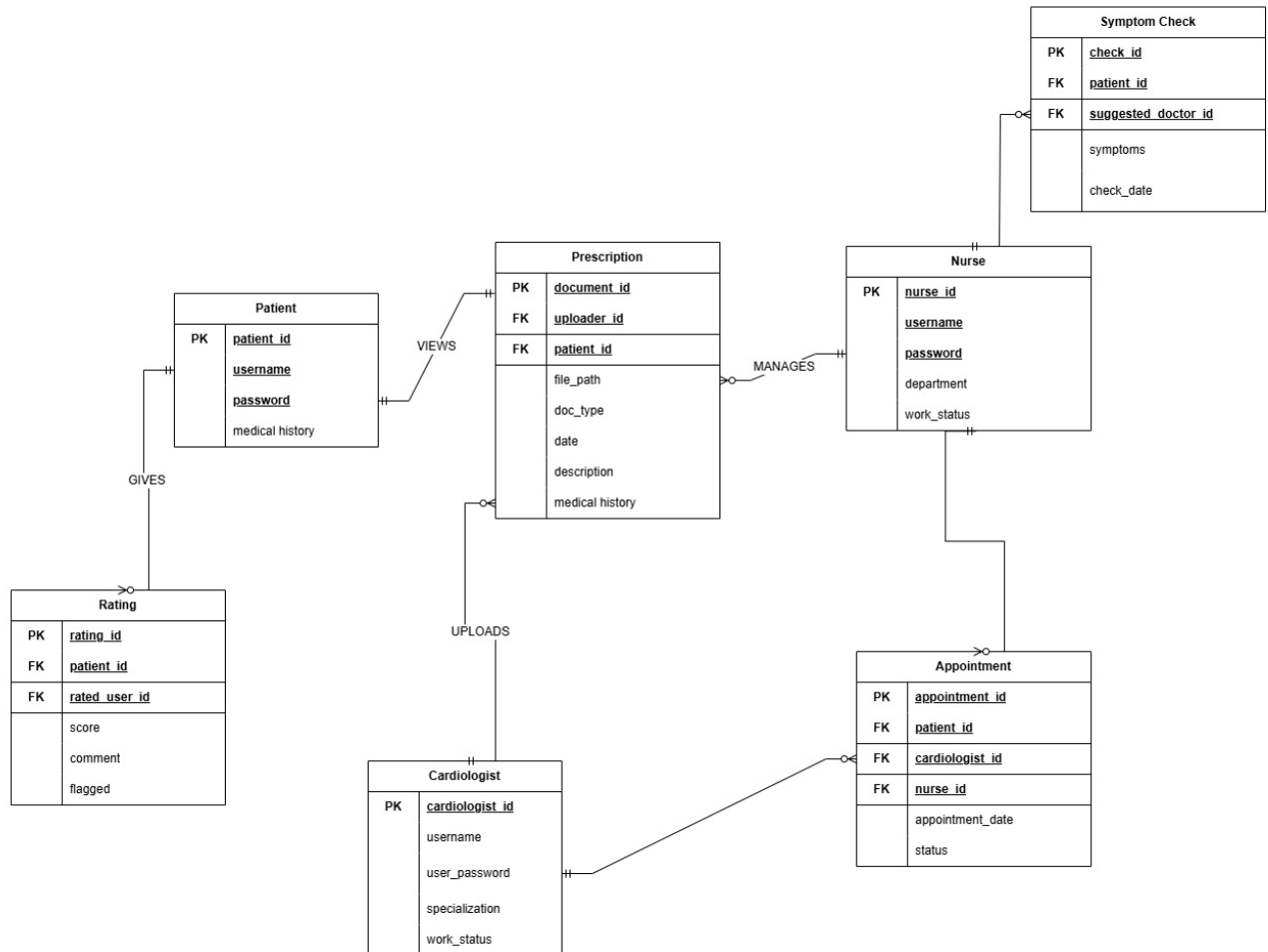


Figure 4.3.3 Entity-Relationship Diagram

4.3.4: Class Diagram

This system has several classes as demonstrated below. The figure below represents the attributes and methods of the various classes. The classes include user, patient, cardiologists, nurse, admin, rating, document, appointment and symptom checker

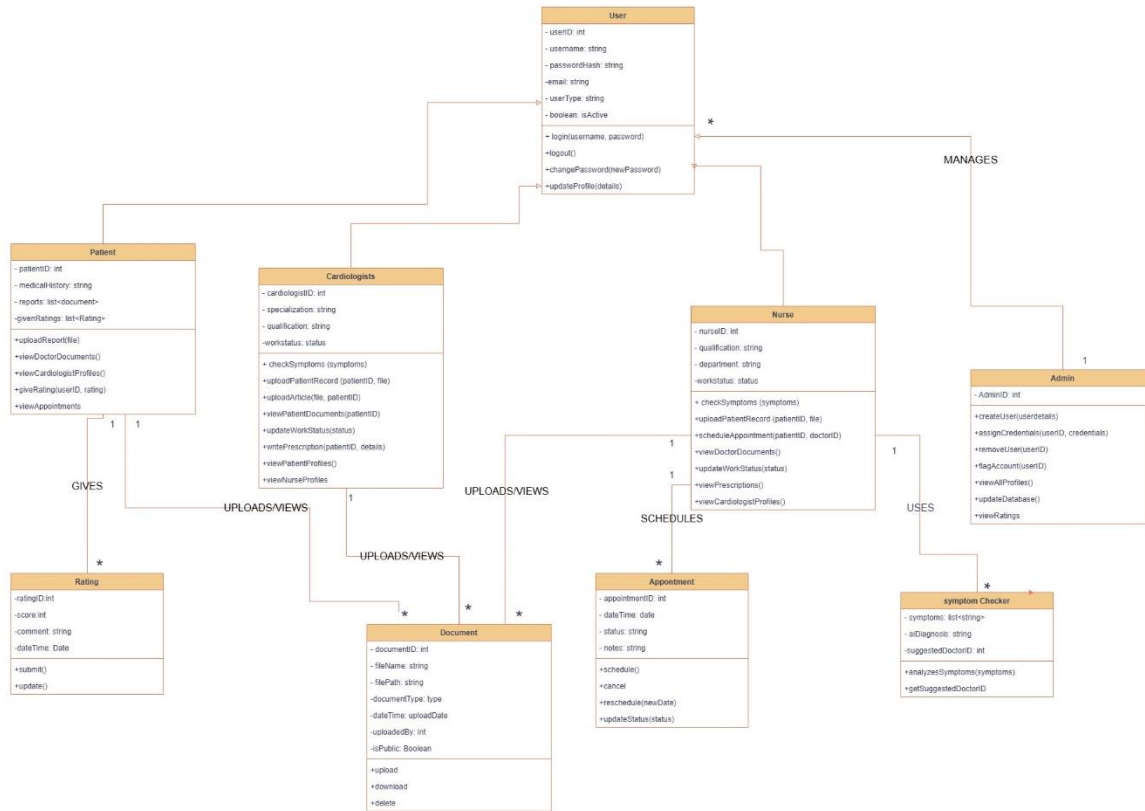


Figure 4.3.4 Class Diagram

4.3.5: Database Schema

This is a blueprint of how the design of the database will look like. It is made up of various tables that make up the system's database together with the different keys and attributes in the relations.

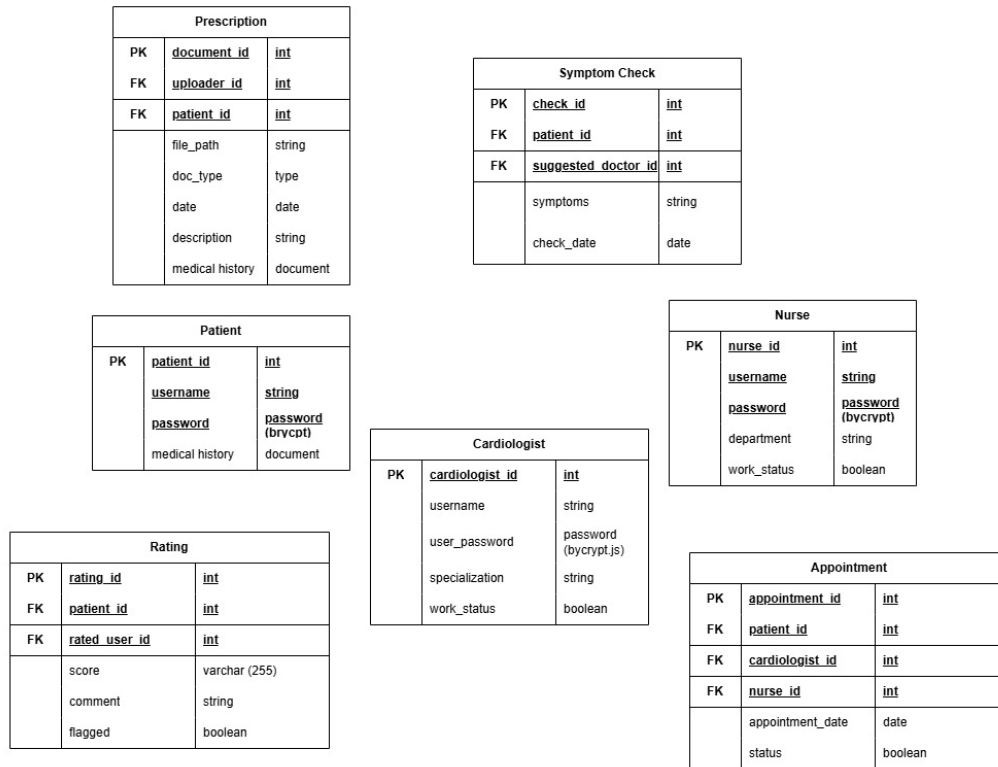


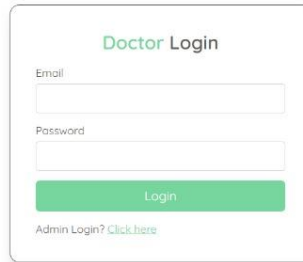
Figure 4.3.5 Database Schema

4.4. Wireframes

The wireframes aid to visualize the system's user interface before development.

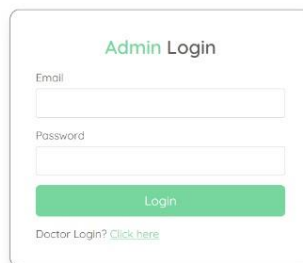
Universal Login and Registration Pages

Both the users and administrators have a uniform login and registration page. They key in their emails and password to gain access. If a patient does not have an account, the user proceeds to create a new account.



The image shows a 'Doctor Login' form. It has a title 'Doctor Login' in green. Below the title are two input fields: 'Email' and 'Password'. Below the 'Password' field is a green 'Login' button. At the bottom, there is a link 'Admin Login? Click here'.

Figure 4.5.5(i): Doctor Login Page



The image shows an 'Admin Login' form. It has a title 'Admin Login' in green. Below the title are two input fields: 'Email' and 'Password'. Below the 'Password' field is a green 'Login' button. At the bottom, there is a link 'Doctor Login? Click here'.

Figure 4.5.5(ii): Admin Login Page

User and Administrator Landing Page

Once the user's credentials have been verified, this page is displayed. They can view new notifications and feedback that are new to the user. The navigation bar entails the name of the user, search bar, page and calendar. The patient can surf for a doctor of their choice by going to the all doctors page and selecting the doctor they need to book an appointment with.

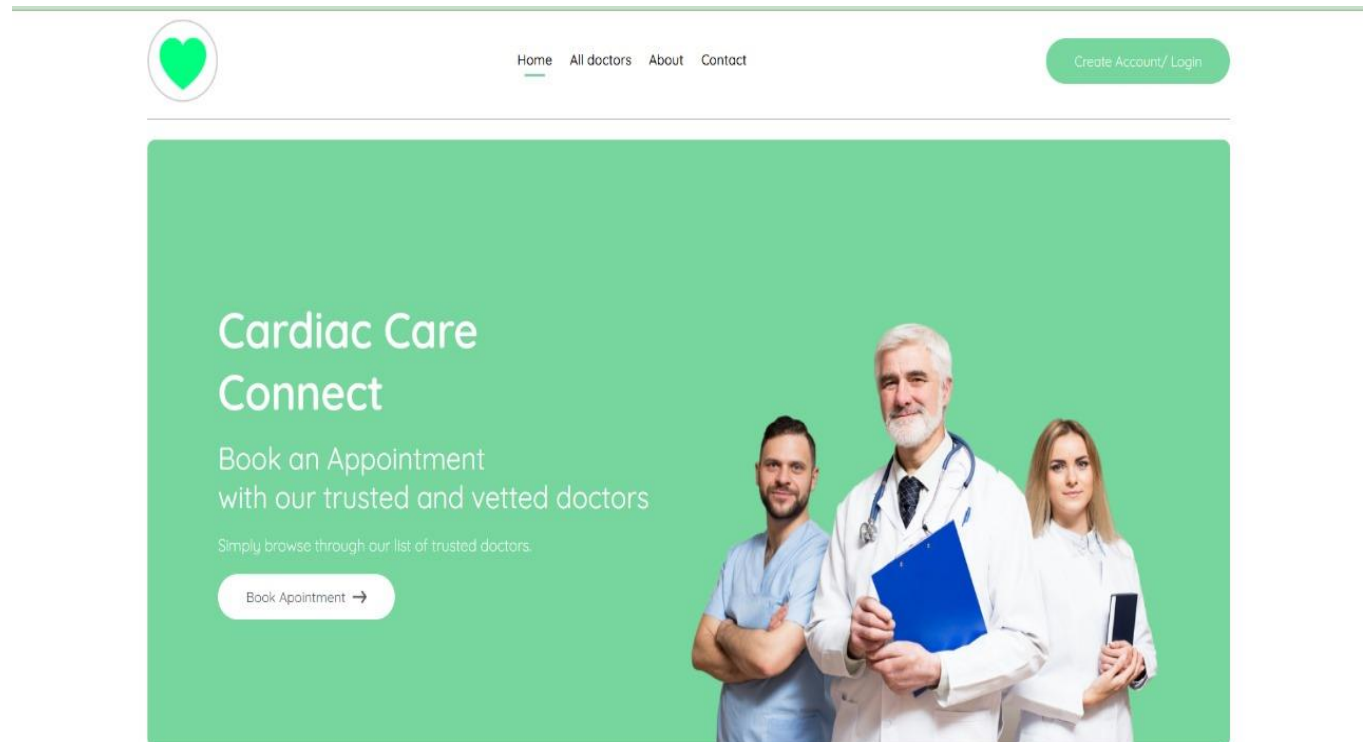


Figure 4.5.5(iii): User Landing Page

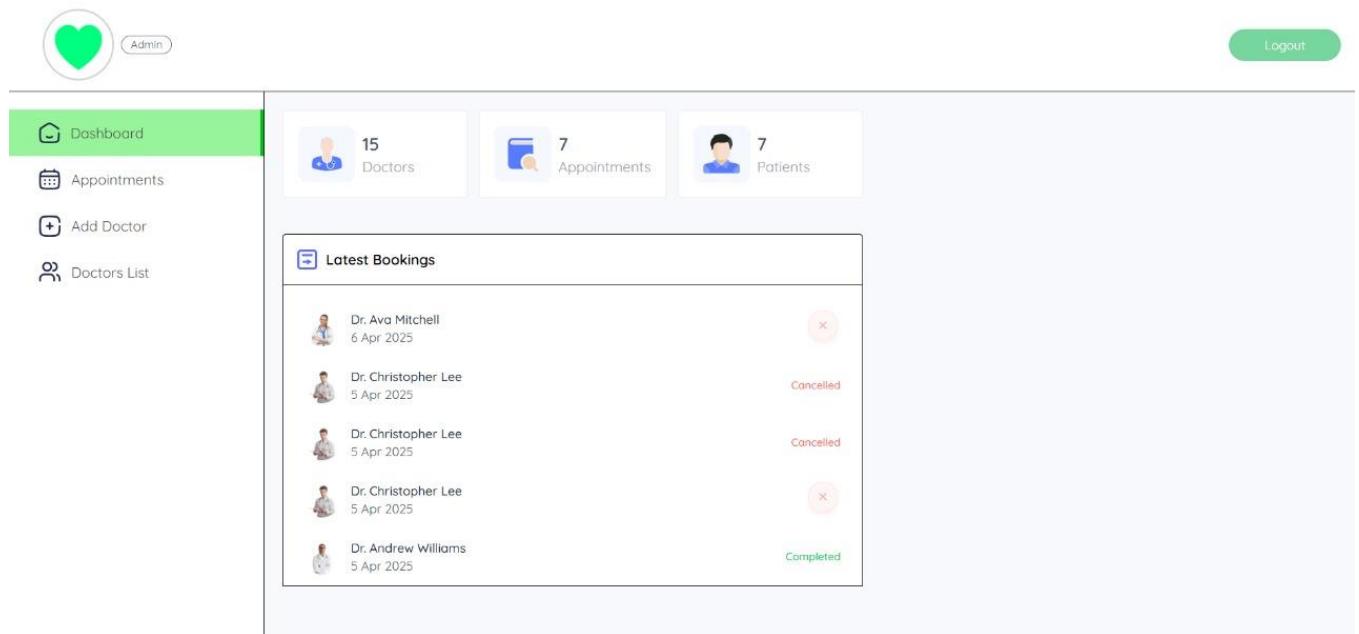


Figure 4.5.5(iv): Admin Landing Page

My Appointments Page

This page displays all the appointments the patient has with their respective doctors at a certain time. The page notifies you if your appointment is still available or if it is cancelled

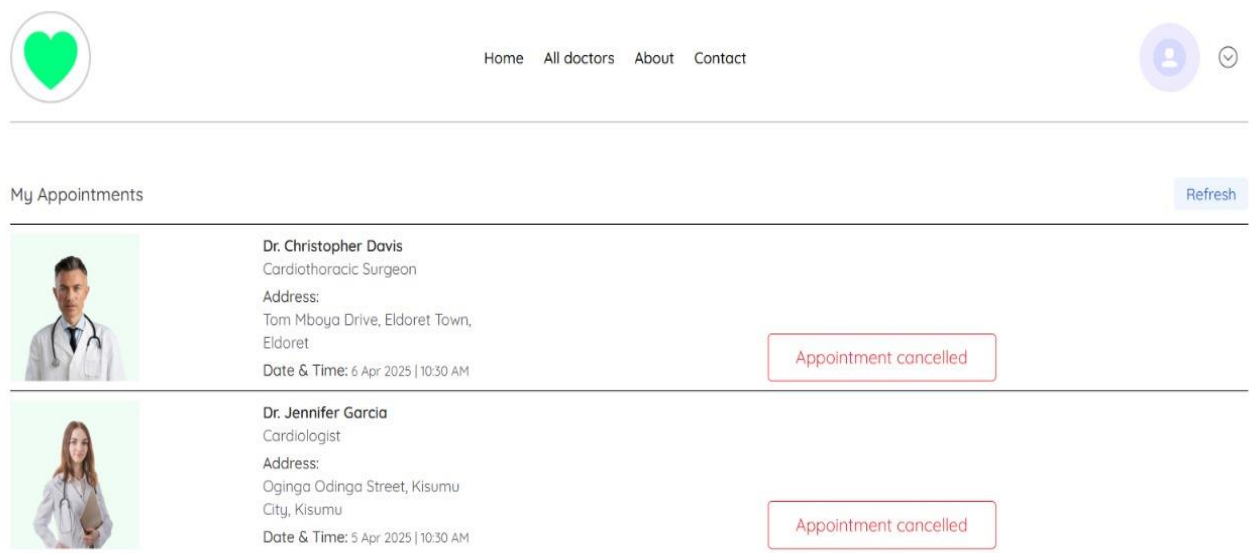


Figure 4.5.5(v): Patient Appointment Page

The Appointment Booking Page

Once the patient has clicked the doctor to book an appointment with the patient gets to see this page where they have booking slots within a certain week.



Dr. Christopher Lee 

MBBS - Cardiac Anesthesiologist 2 Years

About ⓘ

Focuses on perioperative care for heart surgeries, managing pain and sedation while coordinating with surgical teams.

Appointment Fee: Ksh. 3

Booking slots

Fri
4

Sat
5

Sun
6

Mon
7

Tue
8

Wed
9

Thu
10

08:00 pm

08:30 pm

Book an Appointment

Figure 4.5.5(vi): Appointment Booking Page

Chapter Five: System Implementation and Testing

5.1 Introduction

This chapter gives a detailed description of the implementation and testing of the system. It goes on to cover the software and hardware aspects used to make the system efficient and reliable to use.

5.2 Description of the Implementation Environment

This subsection entails the hardware and software specifications of the implementation environments used during the development of the system.

5.2.1 Hardware Specifications

Below is a table that covers all the hardware prerequisites that users need to meet to achieve efficiency of the system performance

Item	Specification	Reason
Processor	Fifth generation processors onwards is recommended for smooth running of Applications. Core i5(8th generation) used for this	To provide enough processing power to handle the application's computational tasks
RAM	8.00 GB onwards	Minimum 8GB RAM for smooth running of the application
Networking	Standard 1Gbps Ethernet connection	This ensures the quick loading of pages of the system and optimizes the users' experience
Storage	Solid State Drive,256GB	For faster data access and responsiveness
Graphics Card	Integrated graphics processor	Enables clear display of the features of the system hence giving the user a comfortable time and experience

Table 5.2.1 Web Application Hardware Specification

5.2.2 Software Specifications

Below shows an overview of the software prerequisites that users need to use the system efficiently.

Item	Specification	Reason
Operating System	Windows 11	For Optimal performance and compatibility.
Web Server	Node js server	Reduces context switching as it allows developers to work with a single language across the stack.
Web Browser	Google Chrome	For optimal running and display of the application.
Database Management System	MongoDB	MongoDB allows for dynamic schemas making it easier to adapt to changing requirements.
React	Javascript Framework	Reusable UI components that speed up development and ensure consistency across the front-end

Table 5.2.2 Web Application Software Specification

5.3 System Implementation

The following subsection describes the modules developed in the system followed by a screenshot of the module.

5.3.1 Homepage and Authentication Module

The homepage is what a normal patient sees when the web application starts. The authentication module would handle the registration and login. It would include features like to either sign in as a patient, forgot password functionality and it would create an account for the new students.

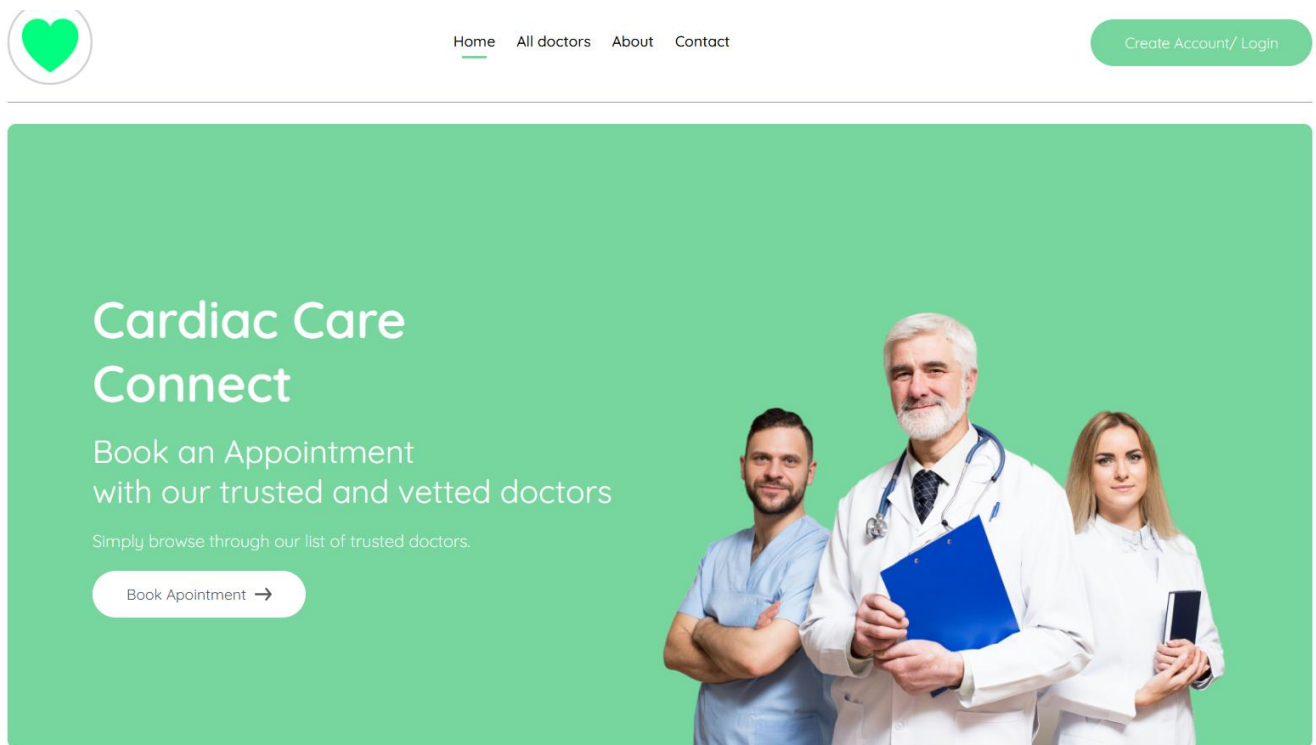


Figure 5.3.1(i): Homepage

Login

Please login to book appointment

Email:

Password

Login

To create a new account? [Click here](#)

Create Account

Please sign up to book appointment

Full name

Email:

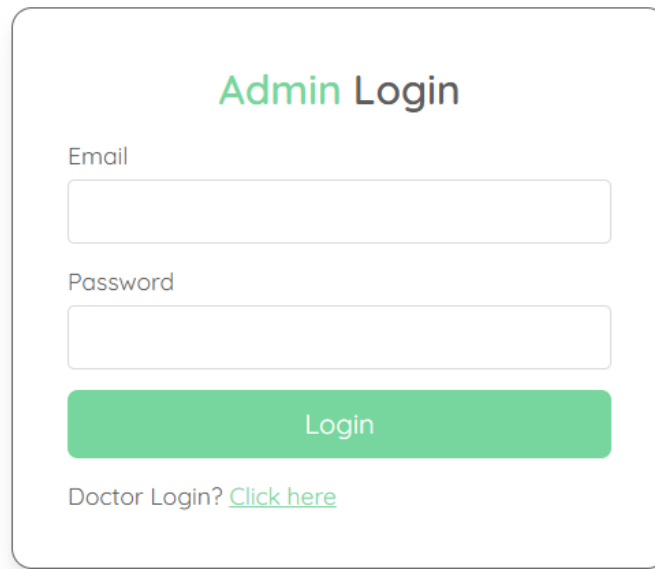
Password

Confirm Password

Sign Up

Already have an account? [Login here](#)

Figure 5.3.1(ii): patient Login Page

The image shows a login form titled "Admin Login" in a green and black font. Below the title are two input fields: "Email" and "Password", both with light gray borders. A green button with the text "Login" is positioned below the password field. At the bottom, there is a link that says "Doctor Login? [Click here](#)".

Admin Login

Email

Password

Login

Doctor Login? [Click here](#)

Figure 5.3.1(iv): Admin Login Page

5.3.2 Patient Module

The patient module would allow patients to book an appointment with a doctor of their choosing. The appointment slots are made available for the doctor according to the actual time during the day i.e. an appointment before the current could not be picked. Which would then lead them to their appointments page.

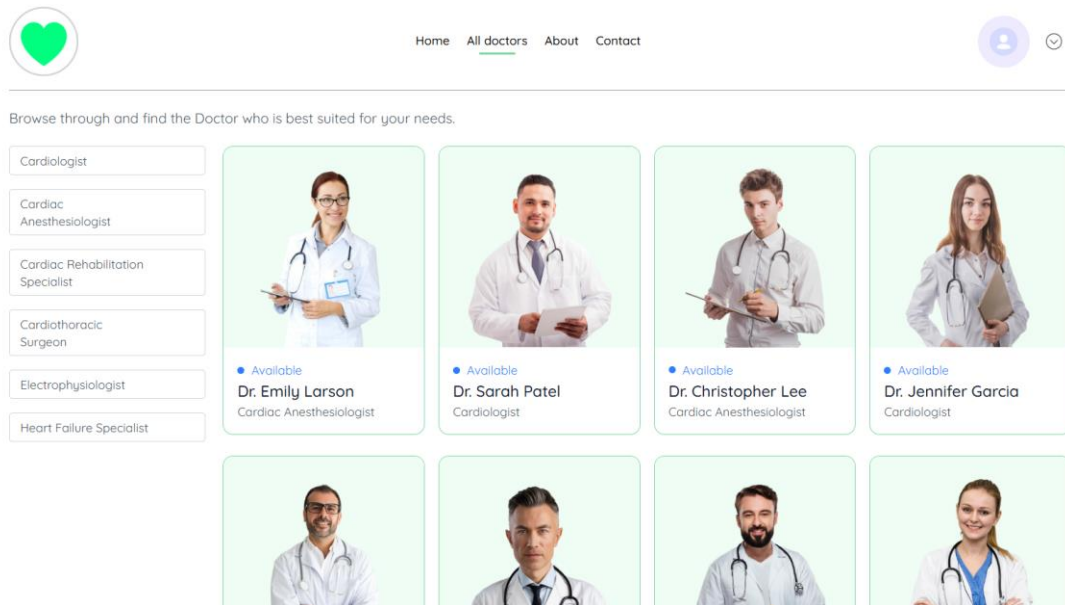
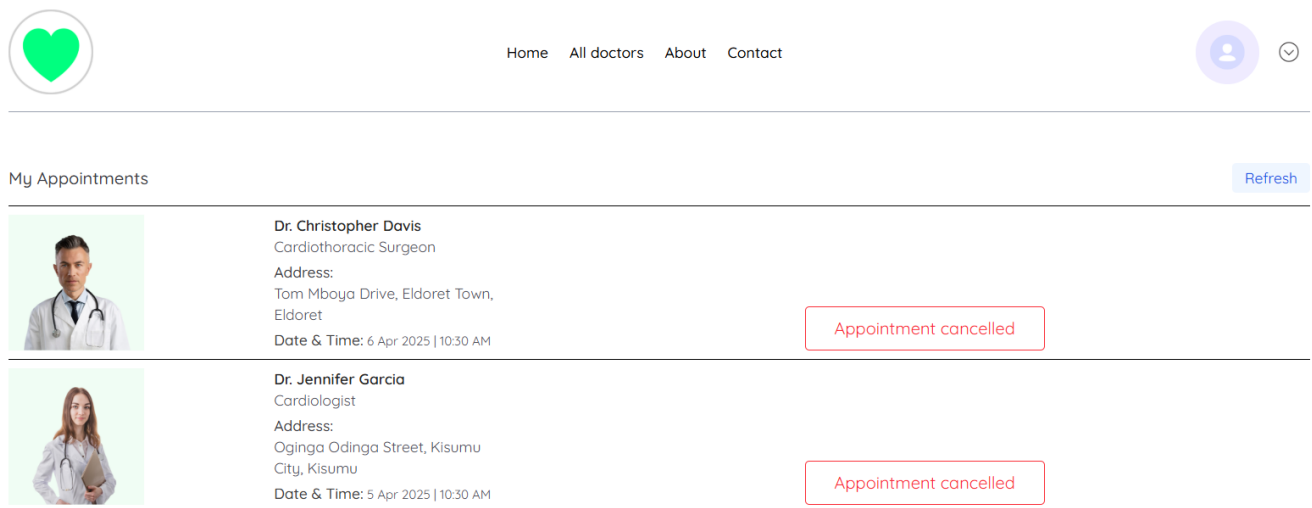
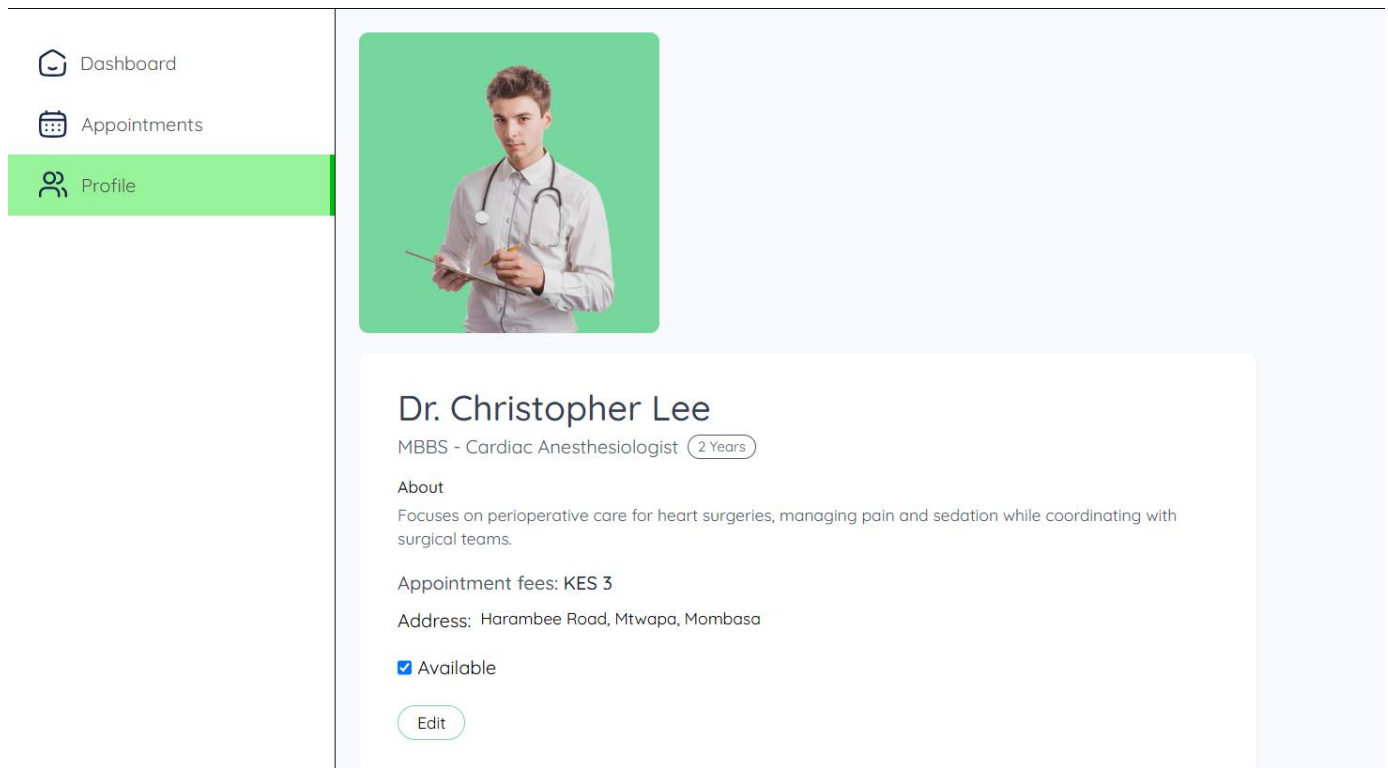


Figure 5.3.2(i): Student Landing Page



5.3.3 Doctor Panel

The doctor panel would allow doctors to manage their quotidian appointments while being able to update their own profile.



5.3.4 Administrator Module

The administrator module would allow viewing of the appointments. Features included as shown below are viewing of the analytics of the administrators.

Dashboard

Appointments

Profile



Dr. Christopher Lee

MBBS - Cardiac Anesthesiologist 2 Years

About

Focuses on perioperative care for heart surgeries, managing pain and sedation while coordinating with surgical teams.

Appointment fees: KES 3

Address: Harambee Road, Mtwapa, Mombasa

☒ Available

Edit

Figure 5.3.4(i): Admin Analytics

Admin

Dashboard

Appointments

Add Doctor

Doctors List

Add Doctor

Upload Doctor Picture

Doctor name:

Name

Doctor Email:

Email

Doctor Password:

Password

Experience

1 Year

fees

fees

Speciality:

Cardiologist

Education

Education

Address:

Address

20

Figure 5.3.4(ii): Adding or Editing of doctors

Dashboard	All Appointments					
Appointments	#	Nurse	Date & Time	Doctor	Fees	Actions
Add Doctor	1	Rick Morty	5 Apr 2025, 10:30 AM	Dr. Jennifer Garcia	KES 2	Cancelled
Doctors List	2	Rick Morty	6 Apr 2025, 10:30 AM	Dr. Christopher Davis	KES 1	Cancelled
	3	Tony Stark	5 Apr 2025, 10:30 AM	Dr. Andrew Williams	KES 2	Completed
	4	Curtis	5 Apr 2025, 10:30 AM	Dr. Christopher Lee	KES 3	
	5	Curtis	5 Apr 2025, 11:30 AM	Dr. Christopher Lee	KES 3	Cancelled
	6	Curtis	5 Apr 2025, 11:30 AM	Dr. Christopher Lee	KES 3	Cancelled
	7	Curtis	6 Apr 2025, 10:00 AM	Dr. Ava Mitchell	KES 3	
	8	Rick Morty	6 Apr 2025, 10:30 AM	Dr. Christopher Lee	KES 3	

Figure 5.3.4(iii): Managing of appointments

5.4 System Testing

An essential phase is system testing. This refers to a procedure used to evaluate the quality, accuracy, and completeness of generated software. It consists of a series of actions carried out with the goal of identifying software bugs so that they can be fixed before the product is made available to end users.

5.4.1 Testing Paradigm

Unit and integration testing were used in this project. Unit tests are used where individual components are tested separately and integration tests are used where whole modules are integrated and tested as a single entity.

5.4.2 Testing Results

This subsection entails the testing outcomes of the web application that was used to demonstrate the model developed in this project.

5.4.2.1 Authentication module

The authentication module was done the following tests to it to check if different usernames and passwords in various combinations would display invalid credentials.

Test Case ID	Related Requirement	Inspection Check	Precondition	Test Data	Result
TC01	The system should allow students to register	Can a student register when they have entered their details in the form?	A user must have entered valid inputs in the registration form	User credentials	Pass
TC02	The system should allow students to log in	Can a student log in once they have filled details in the login form?	A user must be registered by having signed up	User credentials	Pass
TC03	Invalid email input when logging in	What happens when a student inputs a wrong email when logging in?	A user must be registered by signing up and have entered the correct password	Email credentials	System outputs 'Please include an @ in the email address.'
TC04	Invalid password input when logging in	What happens when a user inputs a wrong password when logging in?	A user must be registered by signing up and have entered the correct username	Password credential	System outputs 'Invalid user credentials'
TC05	No input when signing up and logging	What happens when a user makes no	No pre-existing condition	User Credential	System outputs

	in	inputs when logging in or signing up?			'Fill in the required field'
--	----	---------------------------------------	--	--	------------------------------

5.4.2.2 Patient Module

All the functionalities within the user's dashboard once they have logged in were tested manually.

Test Case ID	Related Requirement	Inspection Check	Precondition	Test Data	Result
TC001	Patients can book an appointment	Are the appointment slots responsive to other bookings	They have to be logged in	User capabilities	Pass
TC002	Patients can view their own appointments	Doctors slots that are already booked moved from available slots	No existing precondition	User capabilities	Pass
TC003	Patients view their own details	Patients can edit their own credentials	User must be logged in	User capabilities	Pass

5.4.2.3 Admin Module

Admin functionalities such as editing, viewing and deleting users were tested.

Test Case ID	Related Requirement	Inspection Check	Pre-condition	Test Data	Result
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TC01	Admin can view all user details	Can admin view all user details?	Admin is logged in	Admin capabilities	Pass
TC02	Admin can edit user details	Can admin edit user details?	Admin is logged in	Admin capabilities	Pass
TC03	Admin can delete users	Can admin delete users?	Admin is logged in	Admin capabilities	Pass
TC04	Admin can view all the analytics	Can admin view the line chart and summary of the recent students?	Admin is logged in	Admin capabilities	Pass
TC05	Admin can add events of the club	Can admin add the events of the club?	Admin is logged in	Admin capabilities	Pass
TC06	Admin can assign tasks to a student	Can admin assign tasks to a student?	Admin is logged in	Admin capabilities	Pass
TC07	Admin can view the tasks assigned to students	Can admin view the tasks?	Admin is logged in	Admin capabilities	Pass

Chapter 6: Conclusions, Recommendations and Future works

6.1 Conclusions

The Cardiac Care Connect project successfully achieved its primary objective of developing a personalized and user-friendly patient-centered communication system. It addressed the challenges faced by patients when it comes to attentive medical care by integrating intuitive tools. The project effectively identified and tackled the limitations of the existing solutions that have attempted to ameliorate the dearth of patient-centered communication, emphasizing the need for better customization and real-time interaction features. The developed web-based Cardiac Care Information System enhanced user engagement with minimal learning curves by creating a clear and simple user experience that made navigation and interaction simple. The introduction of the relevant color schemes on the web application made it appear pleasing to the patients and administrators. Effective data management strategies were also put into place to improve user customization while ensuring data security and privacy were protected by strong authentication protocols. To sum up, the Cardiac Care Information system has successfully transformed operations by providing a personalized, simplified, and engaging platform.

6.2 Recommendations

The Cardiac Care Connect application is recommended to any health key user looking for a customized and idiosyncratically useful information system designed to curb the needs of patients. Healthcare organizations that have an implicit need for better communication with their patients would find it very helpful. The program needs to run on machines with at least 4 GB of RAM, any current web browser, and a steady internet connection to function at its best.

To guarantee that the system functions as intended, it is also advised to host the application on a robust server with adequate RAM and a powerful CPU to handle multiple user requests at once. Finally, improving the user interface to make it more intuitive and user-friendly can enhance the user experience even more.

6.3 Future Works

While we have successfully met our primary goals with the Cardiac Care Connect project, there are several promising directions for future development. One exciting avenue is to expand multilingual support, allowing us to connect with a more diverse group of students and staff. Additionally, by incorporating social media features, we can enhance the sharing of club experiences and foster a greater sense of community. Offline capabilities are another area worth exploring, as they would make the application more accessible even when users are not connected to the internet.

Making the application mobile-friendly is a top priority for us, as it would greatly improve usability and convenience for students who are always on the move. In the coming updates, we aim to enrich the application's functionalities, offer more value to our users, and delve into new research opportunities to further enhance the club management experience. Another key focus will be integrating advanced analytics tools, which will help us analyze club performance and provide personalized recommendations. Ultimately, our goal is to transform the Strathmore University Clubs and Societies application into an essential resource for effective club management and meaningful community engagement.

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and documentation processes are followed

3.4.3 System Documentation

A document holding all relevant information about the Web based Cardiac Care Information System and its development shall be provided.

REFERENCES

Detect Text

Upload File

10,046/15,000 Characters

Check 125,000 characters, [Upgrade Here](#)

Your Text is Human written



Different methodologies have been analyzed and recommended in this section; data types have been defined in accordance with the means of data collection that will be used for the data. Analysis and Design diagrams have been drawn to illustrate the best approach to the system's instantiation.

3.2 Research Paradigm

For this particular project, Object Oriented Analysis and Design Method (OOADM) will serve the needs of the Information system design best compared to Structured System Analysis and Design Method (SSADM). This conclusion is due to the following reasons:



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